

HNBR SYNTHESIS, HYDROGENATION, AND DEGRADATION PROCESS LANDSCAPE FROM SELECTED EP/US PATENTS

1. ABSTRACT

This report synthesizes nine primary EP/US patents centered on hydrogenated nitrile butadiene rubber (HNBR) and closely related manufacturing routes. The selected landscape spans direct hydrogenation of NBR to HNBR, continuous hydrogenation in multistage reactors, low-molecular-weight HNBR formation by simultaneous metathesis/hydrogenation, and metathetic degradation of nitrile rubbers as a precursor step to lower-viscosity HNBR. Across the set, the target material is consistently defined by hydrogenation of the diene-derived unsaturation in nitrile rubber backbones, with the requested target attribute being the degree of hydrogenation or residual double-bond content. Several patents emphasize catalyst systems based on ruthenium, rhodium, palladium, osmium, or Grubbs/Hoveyda/Grela-type metathesis complexes, while others focus on process intensification through spinning disk reactors or multistage agitated continuous reactors. Organic solvents dominate the disclosed media, especially monochlorobenzene and chlorobenzene, although one patent uses aqueous emulsion polymerization to make the NBR precursor before coagulation and drying. Water is therefore present in some patents, but usually as a polymerization medium for precursor NBR or as workup/coagulation/washing rather than as the HNBR hydrogenation medium itself. Formulation/process trends include low catalyst loadings in ppm or phr, hydrogen pressures from a few bar to tens of bar, temperatures commonly around 100–140°C, and reaction times ranging from minutes in continuous systems to hours in batch systems. Several patents report residual double-bond levels, hydrogenation percentages, molecular-weight reduction, and improved aging or processability as the key performance outcomes. Overall, the portfolio shows two dominant technical directions: catalyst/process optimization for efficient hydrogenation of NBR to HNBR, and precursor degradation/metathesis strategies to access lower-molecular-weight HNBR with narrower distributions and improved handling.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: ARLANXEO Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses hydrogenation of at least partially unsaturated nitrile rubber (NBR) in solution to produce HNBR, using ruthenium, palladium, or rhodium compounds as hydrogenation catalysts. The focus is on achieving HNBR with improved ageing properties while avoiding a separate catalyst-removal step after hydrogenation. - NBR composition: at least 40% to 90% by weight conjugated diene — Claim 1 - NBR composition: at least 10% to 60% by weight α,β -unsaturated nitrile — Claim 1 - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200°C — Claim 1 - Hydrogenation pressure: 700000 pascals to 15000000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - Ageing test: 4 days at 140°C — Description - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: solution polymerization / solution hydrogenation in NBR solution

degree of hydrogenation - Not disclosed in extracted evidence

Relevant Experimental Evidence - No worked examples were present; the patent relies on claims and description. - The disclosure demonstrates that low noble-metal loadings can still produce HNBR with improved ageing properties. - The patent emphasizes catalyst retention/removal avoidance rather than a specific example-based conversion profile.

Technical Relevance This patent is directly relevant because it is explicitly about producing HNBR from NBR by hydrogenation. It provides catalyst loading and process windows that define practical HNBR synthesis conditions and links them to ageing performance.

Patent 2

- Patent Number: US7666950B2

- Patent Title: Process for preparing hydrogenated nitrile rubbers
- Assignee: Lanxess Deutschland GmbH
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses hydrogenation of nitrile rubber in a spinning disk reactor using homogeneous hydrogenation catalysts, especially rhodium- or ruthenium-containing catalysts, to prepare HNBR. The process is designed to reduce hydrogen pressure requirements and improve manufacturing efficiency while controlling residual double bonds. - Hydrogen pressure: 3.5 bar to 100 bar — Claim 1 - Catalyst loading: 0.36 phr — Example 1, Table 1 - Catalyst loading: 0.60 phr — Example 3, Table 1 - Metal content: 65.6 ppm — Example 1, Table 1 - Metal content: 91.3 ppm — Example 1, Table 1 - Disk temperature: 135°C — Example 1, Table 1 - Disk temperature: 150°C — Example 3, Table 1 - Reaction time: 240 min — Example 1, Table 1 - Solution concentration: 4% b.w. Perbunan T 3429 in MCB — Example 1 - Disk rotational speed: 1000 rpm — Example 1 - Solution circulation: 2 mL/s — Example 1 - Brookfield viscosity test temperature: 22°C +/- 1°C — Example 1 - Brookfield viscometer speed: 60 rpm — Example 1 - Residual double bonds (RDB): 6.0% — Example 1 - Residual double bonds (RDB): 25.4% — Example 2 - Residual double bonds (RDB): 0.9% — Example 3 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: RDB 6.0%, 25.4%, and 0.9% in Examples 1-3; corresponding hydrogenation completeness varies by example

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: organic solvent solution hydrogenation in monochlorobenzene

degree of hydrogenation - RDB 6.0%, 25.4%, and 0.9% in Examples 1-3; corresponding hydrogenation completeness varies by example

Relevant Experimental Evidence - Example 1: HNBR was prepared in a spinning disk reactor at 10 bar H₂, 135°C, and 240 min using 0.36 phr catalyst; the product had 6.0% remaining double bonds. - Example 2: Lower hydrogen pressure (3.5 bar) at the same catalyst loading gave 25.4% remaining double bonds, showing pressure sensitivity. - Example 3: Higher catalyst loading (0.60 phr) and 150°C gave 0.9% remaining double bonds, demonstrating more complete hydrogenation.

Technical Relevance This patent is highly relevant because it directly prepares HNBR and quantifies residual unsaturation after hydrogenation. It also shows how reactor design, pressure, temperature, and catalyst loading affect the degree of hydrogenation.

Patent 3

- Patent Number: EP2473281B1

- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: LANXESS Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses simultaneous hydrogenation and metathesis of nitrile rubber in solution to produce low-molecular-weight HNBR. Ruthenium or osmium catalysts of Grubbs/Hoveyda-type are used, optionally with co-olefin, to reduce molecular weight while hydrogenating the polymer in a single step. - Catalyst amount: 0.041 phr — Example 1, Table 1 - Catalyst amount: 0.045 phr — Example 1, Table 1 - Catalyst amount: 0.060 phr — Example 1, Table 1 - Metal content: 65.6 ppm — Example 1, Table 1 - Metal content: 91.3 ppm — Example 1, Table 1 - Reaction time: 0 to 240 min — Example 1, Table 2 - RDB at 60 min: 2.5% — Example 1, Table 2 - RDB at 60 min: 7.0% — Example 1, Table 2 - RDB at 60 min: 15.4% — Example 1, Table 2 - RDB at 240 min: 0.4% — Example 1, Table 2 - RDB at 240 min: 3.5% — Example 1, Table 2 - RDB at 240 min: 0.7% — Example 1, Table 2 - Mw: 113371 g/mol — Example 1, Table 3 - Mw: 133758 g/mol — Example 1, Table 3 - Mw: 195304 g/mol — Example 1, Table 3 - Mn: 48352 g/mol — Example 1, Table 3 - Mn: 54595 g/mol — Example 1, Table 3 - Mn: 64130 g/mol — Example 1, Table 3 - PDI: 2.34 — Example 1, Table 3 - PDI: 2.45 — Example 1, Table 3 - PDI: 3.05 — Example 1, Table 3 - Mooney viscosity: 33.3 — Example 1, Table 3 - Mooney viscosity: 41.3 — Example 1, Table 3 - Mooney viscosity: 69.3 — Example 1, Table 3 - Acrylonitrile content: 34% by weight — Example 1 - Residual moisture content: <0.5% by weight — Example 1 - Temperature: 130°C — Example 1 - Pressure: 85 bar — Example 1 - Temperature: 138°C — Example 1 - NBR solution concentration: 12% — Example 1 - NBR mass: 518 g — Example 1 - Monochlorobenzene: 4300 g — Example 1 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: RDB reduced to 0.4% in HNBR 1, 3.5% in HNBR 2, and 0.7% in HNBR 3 after 240 min

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: organic solvent solution hydrogenation/metathesis in monochlorobenzene

degree of hydrogenation - RDB reduced to 0.4% in HNBR 1, 3.5% in HNBR 2, and 0.7% in HNBR 3 after 240 min

Relevant Experimental Evidence - Example 1: Hoveyda, Arlt, and Wilkinson catalysts were compared for simultaneous hydrogenation/metathesis; Hoveyda gave the fastest hydrogenation and the lowest Mw after 240 min. - Example 1: At equal metal content, the Hoveyda catalyst reduced RDB to 0.4% by 240 min, versus 0.7% for Wilkinson's catalyst. - Example 1: The process produced low-molecular-weight HNBR with Mw as low as 113371 g/mol and Mooney viscosity 33.3.

Technical Relevance This patent is directly relevant because it produces HNBR while simultaneously lowering molecular weight, a key HNBR manufacturing route. It also provides explicit residual unsaturation data that reflect the degree of hydrogenation achieved in the target material.

Patent 4

- Patent Number: US9079979B2
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Lanxess Deutschland GmbH
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen and a ruthenium/osmium catalyst, producing lower-molecular-weight HNBR. The process is carried out in monochlorobenzene solution and compares Grubbs III, Grubbs II, and Wilkinson-type catalysts. - Hydrogen pressure: 7 bar — Example 1, precursor degassing - Hydrogen pressure: 85 bar — Example 1 - Temperature: 130°C — Example 1 - Temperature: 138°C — Example 1 - Reaction time: 0 to 240 min — Example 1, Table 2 - Catalyst amount: 0.057 phr — Example 1, Table 1 - Catalyst amount: 0.080 phr — Example 1, Table 1 - Catalyst amount: 0.060 phr — Example 1, Table 1 - Catalyst amount: 0.055 phr — Example 1, Table 1 - Metal content: 65.6 ppm — Example 1, Table 1 - Metal content: 91.3 ppm — Example 1, Table 1 - Acrylonitrile content: 34% by weight — Example 1 - Residual moisture content: <0.5% by weight — Example 1 - NBR mass: 518 g — Example 1 - Monochlorobenzene: 4300 g — Example 1 - RDB at 60 min: 2.0% — Example 1, Table 2 - RDB at 60 min: 0.7% — Example 1, Table 2 - RDB at 60 min: 15.4% — Example 1, Table 2 - RDB at 60 min: 2.9% — Example 1, Table 2 - RDB at 120 min: 0.4% — Example 1, Table 2 - RDB at 120 min: 0.5% — Example 1, Table 2 - RDB at 120 min: 3.1% — Example 1, Table 2 - RDB at 120 min: 0.6% — Example 1, Table 2 - Mw: 191541 g/mol — Example 1, Table 3 - Mw: 135772 g/mol — Example 1, Table 3 - Mw: 195304 g/mol — Example 1, Table 3 - Mw: 144212 g/mol — Example 1, Table 3 - Mn: 68643 g/mol — Example 1, Table 3 - Mn: 58259 g/mol — Example 1, Table 3 - Mn: 64130 g/mol — Example 1, Table 3 - Mn: 59851 g/mol — Example 1, Table 3 - PDI: 2.79 — Example 1, Table 3 - PDI: 2.33 — Example 1, Table 3 - PDI: 3.05 — Example 1, Table 3 - PDI: 2.45 — Example 1, Table 3 - Mooney viscosity: 38.2 — Example 1, Table 3 - Mooney viscosity: 20.2 — Example 1, Table 3 - Mooney viscosity: 69.3 — Example 1, Table 3 - Mooney viscosity: 32.6 — Example 1, Table 3 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: RDB reduced to 0.3% by 180 min and 0.4% by 240 min for HNBR 1; other HNBRs show 0.5%, 3.1%, 0.6%, etc. depending on catalyst

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: organic solvent solution hydrogenation/metathesis in monochlorobenzene

degree of hydrogenation - RDB reduced to 0.3% by 180 min and 0.4% by 240 min for HNBR 1; other HNBRs show 0.5%, 3.1%, 0.6%, etc. depending on catalyst

Relevant Experimental Evidence - Example 1: Grubbs III catalyst gave faster hydrogenation than Grubbs II and Wilkinson's catalyst at equal metal content. - Example 1: HNBR 1 reached 0.3% RDB by 180 min and 0.4% by 240 min, indicating near-complete hydrogenation. - Example 1: HNBR 2 and HNBR 4 showed lower Mooney viscosity than HNBR 3, demonstrating catalyst-dependent processability.

Technical Relevance This patent is directly relevant because it produces HNBR by simultaneous hydrogenation/metathesis and explicitly tracks residual double bonds. It is important for understanding how catalyst choice affects the degree of hydrogenation and final HNBR properties.

Patent 5

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Zannan Scitech Co., Ltd.
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses synthesis of ruthenium metathesis catalysts and their use in metathesis reactions such as ROMP, RCM, and also depolymerization/hydrogenation-related rubber transformations. It is primarily a catalyst patent rather than a direct HNBR preparation patent, but it explicitly mentions modified nitrile butadiene rubber and depolymerized HNBR as application contexts. - Ligand preparation scale: 5.0 mmol — Example 1 - Ligand preparation scale: 1.0 mmol — Example 1 - Reaction temperature: 85°C — Example 1 - Reaction temperature: 20-25°C — Example 1 - Reaction time: 30-60 min — Example 1 - Yield: 86% — Example 1 - Yield: 4% — Example 1 - Yield: 91% — Example 2 - Yield: 29% — Example 2 - Yield: 54% — Example 3 - Yield: 5% — Example 3 - Yield: 62% — Example 4 - Yield: 32% — Example 4 - Yield: 35% — Example 5 - Yield: 14% — Example 6 - Yield: 56% — Example 7 - Yield: 8% — Example 7 - Yield: 32% — Example 8 - Yield: 42% — Example 8 - Yield: 93% — Example 9 - Yield: 48% — Example 9 - Yield: 83% — Example 10 - Yield: 68% — Example 10 - Yield: 49% — Example 11 - Yield: 52% — Example 11 - Yield: 45% — Example 12 - Yield: 64% — Example 12 - Yield: 67% — Example 13 - Yield: 26% — Example 13 - Yield: 46% — Example 14 - Yield: 38% — Example 14 - Yield: 44% — Example 15 - Yield: 14% — Example 15 - Yield: 51% — Example 16 - Yield: 85% — Example 16 - Yield: 38% — Example 17 - Yield: 23% — Example 17 - Yield: 28% — Example 18 - Yield: 32% — Example 18 - Yield: 81% —

Example 19 - Yield: 10% — Example 19 - DME: 10 mL — Example 1 - Deionized water: 3 mL — Example 1 - K₂CO₃: 1.5 eq — Example 1 - Pd(PPh₃)₄: 2% — Example 1 - DCM: 15 mL — Example 1 - Pentane/DCM gradient: 2/1 to DCM — Example 1 - SM-3a: 5.0 mmol — Example 1 - SM-3b: 5.0 mmol — Example 2 - Ru complex 1b: 1.0 mmol — Example 1 - Ru complex 1a: 1.0 mmol — Example 2 - Medium & Water Role: Water is not the polymerization medium; it appears only in a ligand-synthesis step and downstream workup, while the core reactions are carried out in organic solvent (DCM/DME) under inert atmosphere. - degree of hydrogenation: Not disclosed in extracted evidence

Medium & Water Role - Water is not the polymerization medium; it appears only in a ligand-synthesis step and downstream workup, while the core reactions are carried out in organic solvent (DCM/DME) under inert atmosphere. - Core reaction medium: organic solvent synthesis in DCM/DME; water used only in ligand synthesis/workup

degree of hydrogenation - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Ligand 3a and Ru complex 4a were synthesized; the Ru complex was obtained in 4% yield. - Example 2: Ligand 3b and Ru complex 4b were synthesized; the Ru complex was obtained in 29% yield. - Examples 3-19: A family of Ru complexes 4c-4v was prepared with varying yields, establishing a catalyst library for metathesis applications.

Technical Relevance This patent is only indirectly relevant to HNBR because it discloses metathesis catalysts that can be used in rubber depolymerization and hydrogenation contexts. It does not itself teach HNBR synthesis, but it provides catalyst chemistry that overlaps with HNBR degradation routes.

Patent 6

- Patent Number: US9023914B2
- Patent Title: Nitrile rubbers and production thereof in organic solvents
- Assignee: Lanxess Deutschland GmbH
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses free-radical polymerization of nitrile rubbers in organic solvents with specific RAFT modifiers, followed optionally by hydrogenation to HNBR. The hydrogenation is described as advantageously taking place in organic solvent, and the examples focus on NBR precursor synthesis and precursor properties rather than direct HNBR hydrogenation data. - Acrylonitrile: 37 phm — Example 1 - 1,3-butadiene: 63 phm — Example 1 - Dimethylacetamide: 51 phm — Example 1 - Vazo 88: 0.31

phm — Example 1 - DoPAT: 0.22 phm — Example 1 - Polymerization temperature: 100°C — Example 1 - Polymerization time: 9 h — Example 1 - Final conversion: 50% — Example 1 - Acrylonitrile: 30 phm — Example 6 - Hydroxyethyl acrylate: 20 phm — Example 6 - Polymerization time: 6 h — Example 6 - Final conversion: 39% — Example 6 - Acrylonitrile: 38 phm — Example 9-16 - 1,3-butadiene: 62 phm — Example 9-16 - Polymerization temperature: 100°C — Example 9-16 - Polymerization time: 22 h — Example 9-16 - Final conversion: 37% to 43% — Example 9-16 - Acrylonitrile: 38 phm — Example 17-23 - 1,3-butadiene: 62 phm — Example 17-23 - HDODA: 6.63 to 15.9 mmol — Example 17-23 - HEA: 159 mmol — Example 21 - Polymerization temperature: 100°C — Example 17-23 - Polymerization time: 9 h — Example 17-23 - Final conversion: 36% to 51% — Example 17-23 - Water: 200 phm — Comparative Examples A and B - Edenor HTiCT, Na/K salt: 2.0 phm — Comparative Examples A and B - Trigonox NT50: 0.020 phm — Comparative Examples A and B - Activator solution: 0.022 phm — Comparative Examples A and B - Polymerization temperature: 13°C — Comparative Examples A and B - Polymerization time: 8.5 h and 9.5 h — Comparative Examples A and B - Final conversion: 56% — Comparative Examples A and B - Gel content: 0.4% to 5.0% — Table 3 - Tg: -24.0°C to -19.6°C — Table 3 - Mn: 58,000 to 42,800 g/mol — Table 3 - Mw: 98,000 to 75,300 g/mol — Table 3 - PDI: 1.3 to 1.9 — Table 3 - Medium & Water Role: The inventive NBR examples are carried out in organic solvent solution polymerization, while water appears in the comparative emulsion process as the polymerization medium and later in coagulation/washing/workup. - degree of hydrogenation: Not disclosed in extracted evidence

Medium & Water Role - The inventive NBR examples are carried out in organic solvent solution polymerization, while water appears in the comparative emulsion process as the polymerization medium and later in coagulation/washing/workup. - Core reaction medium: organic solvent solution polymerization; water only in comparative emulsion precursor examples and workup

degree of hydrogenation - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: NBR was prepared in dimethylacetamide at 100°C with DoPAT modifier and 50% conversion, demonstrating organic-solvent solution polymerization. - Example 6: A terpolymer containing hydroxyethyl acrylate was prepared in organic solvent and reached 39% conversion after 6 h. - Examples 17-23: RAFT-modified NBRs were prepared with varying monomer/initiator/modifier levels, giving PDI values as low as 1.7-1.8. - Comparative Examples A and B: Emulsion polymerization used water, emulsifier, and hydroperoxide/activator system, followed by coagulation, washing, and drying.

Technical Relevance This patent is relevant as an HNBR precursor-manufacturing route and because it explicitly states that the resulting nitrile rubbers may be hydrogenated to HNBR. However, the extracted examples mainly characterize NBR precursors and do not directly disclose HNBR hydrogenation outcomes.

Patent 7

- Patent Number: EP2147933B1
- Patent Title: Process for the continuous hydrogenation of carbon-carbon double bonds in an ...

- Assignee: LANXESS Inc.
- Jurisdiction: EP
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses continuous hydrogenation of unsaturated polymers based on conjugated diolefins and other copolymerizable monomers in a multistage agitated reactor. NBR dissolved in monochlorobenzene is fed continuously with hydrogen and an osmium catalyst to produce hydrogenated polymer, including HNBR-type material. - Reactor inner diameter: 0.06 m — Example 1 - Number of stages: 6 — Example 1 - Height of each stage: 0.06 m — Example 1 - Central opening: 5% of whole cross-sectional area — Example 1 - Impeller diameter: 0.03 m — Example 1 - Temperature: 130°C — Example 1 - Catalyst concentration: 100 μ M — Example 1 - Impeller speed: 600 rpm — Example 1 - Pressure: 350 psi (2.41 MPa) — Example 1 - NBR concentration: 2.5 wt.-% — Example 1 - Liquid feed rate: 24 mL/min — Example 1 - Hydrogen gas flow rate: 48 mL/min — Example 1 - Hydrogenation level: 91% — Example 1 - Temperature: 140°C — Example 2 - Pressure: 500 psig (3.45 MPa) — Example 2 - Catalyst concentration: 80 μ M — Example 2 - Impeller speed: 750 rpm — Example 2 - Hydrogenation level: 99.4% — Example 2 - Catalyst concentration: 27 μ M — Example 3 - Hydrogenation level: 61% — Example 3 - NBR concentration: 5 wt.-% — Example 4 - Catalyst concentration: 140 μ M — Example 4 - Hydrogenation level: 99% — Example 4 - Liquid feed rate: 48 mL/min — Example 5 - Hydrogenation level: 85% — Example 5 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: 91%, 99.4%, 61%, 99%, and 85% hydrogenation levels in Examples 1-5

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: organic solvent solution hydrogenation in monochlorobenzene

degree of hydrogenation - 91%, 99.4%, 61%, 99%, and 85% hydrogenation levels in Examples 1-5

Relevant Experimental Evidence - Example 1: Continuous hydrogenation of 2.5 wt.-% NBR in MCB at 130°C and 350 psi gave 91% hydrogenation. - Example 2: Increasing pressure to 500 psig and temperature to 140°C raised hydrogenation to 99.4%. - Example 3: Lower catalyst concentration (27 μ M) reduced hydrogenation to 61%, showing catalyst loading sensitivity. - Example 4: Higher NBR concentration (5 wt.-%) still achieved 99% hydrogenation at 140°C and 500 psig. - Example 5: Doubling liquid feed rate to 48 mL/min reduced hydrogenation to 85%, demonstrating operating flexibility limits.

Technical Relevance This patent is directly relevant because it produces hydrogenated polymer from NBR in a continuous reactor and explicitly reports hydrogenation levels. It is especially useful for understanding how continuous processing conditions control the degree of hydrogenation in

HNBR-type products.

Patent 8

- Patent Number: US8362154B2
- Patent Title: Metathesis of nitrile rubbers in the presence of transition metal complex ...
- Assignee: Lanxess Deutschland GmbH
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses metathetic degradation of nitrile rubber in chlorobenzene solution using ruthenium or osmium catalysts, with 1-hexene as co-olefin in the examples, to lower molecular weight and narrow the molecular weight distribution. Although the main focus is degradation rather than hydrogenation, the patent explicitly references subsequent hydrogenation and HNBR-related products in the background and claims context. - NBR mass: 40 g — General Description - Chlorobenzene: 293.3 g — General Description - 1-Hexene: 0.8 g (2 phr) — General Description - Dissolution time: 10 h — General Description - Catalyst amount: 23 ppm Ru — Example 1.1 - Catalyst amount: 57 ppm Ru — Example 2.1 - Catalyst amount: 22.9 ppm Ru — Example 4.1 - Catalyst amount: 59.5 ppm Ru — Example 5.1 - Catalyst amount: 59.6 ppm Ru — Example 5.2 - Reaction temperature: 23°C — Example tables - Reaction time: 0 to 425 min — Example 1.1 and 1.2 - Reaction time: 0 to 1325 min — Example 2.2 - Mw: 240 kg/mol — Example 1.1 and 1.2 - Mw: 22 kg/mol — Example 1.1 - Mw: 16 kg/mol — Example 1.1 - Mw: 14 kg/mol — Example 1.1 - Mw: 13 kg/mol — Example 1.1 - Mw: 86 kg/mol — Example 1.2 - Mw: 65 kg/mol — Example 1.2 - Mw: 35 kg/mol — Example 1.2 - Mw: 28 kg/mol — Example 1.2 - Mn: 100 kg/mol — Example 1.1 and 1.2 - Mn: 71 kg/mol — Example 1.1 - Mn: 72 kg/mol — Example 1.1 - Mn: 40 kg/mol — Example 1.2 - Mn: 35 kg/mol — Example 1.2 - PDI: 2.4 — Example 1.1 and 1.2 - PDI: 1.57 — Example 1.1 - PDI: 1.33 — Example 1.1 - PDI: 1.40 — Example 1.1 - PDI: 1.30 — Example 1.1 - PDI: 2.13 — Example 1.2 - PDI: 1.87 — Example 1.2 - PDI: 1.88 — Example 1.2 - PDI: 1.68 — Example 1.2 - Acrylonitrile content: 35% by weight — Example section - Mooney viscosity: 34 Mooney units — Example section - Residual moisture content: 1.8% by weight — Example section - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: organic solvent solution metathesis in chlorobenzene/MCB

degree of hydrogenation - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1.1: Hoveyda catalyst at 23 ppm Ru reduced Mw from 240 to 13 kg/mol within 425 min and gave gel-free degraded nitrile rubber. - Example 1.2:

Grubbs II catalyst at the same Ru loading was slower and gave higher Mw values than Hoveyda. - Example 2.1: Hoveyda catalyst at 57 ppm Ru reduced Mw to 36 kg/mol by 90 min and produced gel-free material. - Example 4.1: Grela catalyst outperformed Grubbs II at similar Ru loading, giving lower Mw and gel-free product. - Example 5.1: Grela catalyst at 59.5 ppm Ru achieved Mw 31 kg/mol by 185 min, faster than Grubbs II in Example 5.2.

Technical Relevance This patent is relevant because it addresses nitrile rubber degradation routes that are used to generate lower-molecular-weight feedstocks for HNBR-related materials. It does not directly disclose HNBR hydrogenation degree, but it is important for precursor modification and process efficiency in the HNBR domain.

Patent 9

- Patent Number: US9593168B2
- Patent Title: Process for degrading nitrile rubbers in the presence of catalysts having an ...
- Assignee: Arlanxeo Deutschland Gmbh
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses metathetic degradation of nitrile rubber in chlorobenzene solution using Hoveyda and Grela catalysts, with 1-hexene as co-olefin, to reduce molecular weight and improve catalyst activity relative to Grubbs II. The disclosure is focused on degradation of NBR rather than direct HNBR synthesis, though it is relevant to precursor modification routes in the HNBR field. - NBR mass: 40 g — General Description - Chlorobenzene: 293.3 g — General Description - 1-Hexene: 0.8 g (2 phr) — General Description - Dissolution time: 10 h — General Description - Catalyst amount: 23.8 ppm Ru — Example 2.1 - Catalyst amount: 161.4 ppm Ru — Example 1.1 - Catalyst amount: 22.9 ppm Ru — Example 4.1 - Catalyst amount: 59.5 ppm Ru — Example 5.1 - Catalyst amount: 59.3 ppm Ru — Example 6.1 - Catalyst amount: 59.6 ppm Ru — Example 6.2 - Reaction temperature: 23°C — Example tables - Reaction time: 0 to 425 min — Example 1.1 and 1.2 - Reaction time: 0 to 1325 min — Example 2.2 - Mw: 240 kg/mol — Example 1.1 and 1.2 - Mw: 22 kg/mol — Example 1.1 - Mw: 16 kg/mol — Example 1.1 - Mw: 14 kg/mol — Example 1.1 - Mw: 13 kg/mol — Example 1.1 - Mw: 79 kg/mol — Example 2.1 - Mw: 77 kg/mol — Example 2.1 - Mw: 72 kg/mol — Example 2.1 - Mw: 70 kg/mol — Example 2.1 - Mw: 190 kg/mol — Example 2.2 - Mw: 180 kg/mol — Example 2.2 - Mw: 150 kg/mol — Example 2.2 - Mw: 125 kg/mol — Example 2.2 - Mw: 37 kg/mol — Example 5.1 - Mw: 35 kg/mol — Example 5.1 - Mw: 31 kg/mol — Example 5.1 - Mw: 206 kg/mol — Example 6.2 - Mw: 173 kg/mol — Example 6.2 - Mw: 158 kg/mol — Example 6.2 - Mn: 100 kg/mol — Example 1.1 and 1.2 - Mn: 71 kg/mol — Example 1.1 - Mn: 72 kg/mol — Example 1.1 - Mn: 55 kg/mol — Example 2.1 - Mn: 40 kg/mol — Example 2.1 - Mn: 39 kg/mol — Example 2.1 - Mn: 67 kg/mol — Example 2.2 - Mn: 63 kg/mol — Example 2.2 - Mn: 59 kg/mol — Example 2.2 - Mn: 62 kg/mol — Example 2.2 - Mn: 23 kg/mol — Example 5.1 - Mn: 22 kg/mol — Example 5.1 - Mn: 20

kg/mol — Example 5.1 - Mn: 92 kg/mol — Example 6.2 - Mn: 74 kg/mol — Example 6.2 - Mn: 75 kg/mol — Example 6.2 - PDI: 2.4 — Example 1.1 and 1.2 - PDI: 1.44 — Example 2.1 - PDI: 1.90 — Example 2.1 - PDI: 1.82 — Example 2.1 - PDI: 1.77 — Example 2.1 - PDI: 2.83 — Example 2.2 - PDI: 2.86 — Example 2.2 - PDI: 2.54 — Example 2.2 - PDI: 2.02 — Example 2.2 - PDI: 1.61 — Example 5.1 - PDI: 1.59 — Example 5.1 - PDI: 1.50 — Example 5.1 - PDI: 1.55 — Example 5.1 - PDI: 2.26 — Example 6.1 - PDI: 2.27 — Example 6.1 - PDI: 2.13 — Example 6.1 - PDI: 2.00 — Example 6.1 - Acrylonitrile content: 35% by weight — Example section - Mooney viscosity: 34 Mooney units — Example section - Residual moisture content: 1.8% by weight — Example section - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: organic solvent solution metathesis in chlorobenzene/MCB

degree of hydrogenation - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1.1: Hoveyda catalyst at 161.4 ppm Ru reduced Mw to 13 kg/mol by 425 min and produced gel-free degraded nitrile rubber. - Example 1.2: Grubbs II catalyst at the same Ru loading was slower and retained much higher Mw values. - Example 2.1: Hoveyda catalyst at 23.8 ppm Ru reduced Mw to 70 kg/mol by 425 min and remained gel-free. - Example 5.1: Grela catalyst at 59.5 ppm Ru reduced Mw to 31 kg/mol by 425 min, outperforming Grubbs II. - Example 6.1: Hoveyda catalyst without 1-hexene still outperformed Grubbs II in molecular-weight reduction.

Technical Relevance This patent is relevant to HNBR synthesis strategy because it provides a route for metathetic degradation of nitrile rubber feedstocks, which can be used to tailor precursor molecular weight before hydrogenation. It does not directly disclose HNBR hydrogenation degree, but it is central to precursor engineering in the HNBR domain.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	degree of hydrogenation	Key Findings
EP3827029A1	ARLANXEO Deutschland GmbH	2021	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	No worked examples were present; the patent relies on claims and description.

US7666950B2	Lanxess Deutschland GmbH	2010	No water-related process step disclosed in the extracted evidence.	RDB 6.0%, 25.4%, and 0.9% in Examples 1-3; corresponding hydrogenation completeness varies by example	Example 1: HNBR was prepared in a spinning disk reactor at 10 bar H ₂ , 135°C, and 240 min using 0.36 phr catalyst; the pr
EP2473281B1	LANXESS Deutschland GmbH	2014	No water-related process step disclosed in the extracted evidence.	RDB reduced to 0.4% in HNBR 1, 3.5% in HNBR 2, and 0.7% in HNBR 3 after 240 min	Example 1: Hoveyda, Arlt, and Wilkinson catalysts were compared for simultaneous hydrogenation/metathesis Hoveyda gave

US9079979B2	Lanxess Deutschland GmbH	2015	No water-related process step disclosed in the extracted evidence.	RDB reduced to 0.3% by 180 min and 0.4% by 240 min for HNBR 1; other HNBRs show 0.5%, 3.1%, 0.6%, etc. depending on catalyst	Example 1: Grubbs III catalyst gave faster hydrogenation than Grubbs II and Wilkinson's catalyst at equal metal content.
EP2519350B1	Zannan Scitech Co., Ltd.	2019	Water is not the polymerization medium; it appears only in a ligand-synthesis step and downstream workup, while the core reactions are carried out in organic solvent (DCM/DME) under inert atmosphere.	Not disclosed in extracted evidence	Example 1: Ligand 3a and Ru complex 4a were synthesized; the Ru complex was obtained in 4% yield.

US9023914B2	Lanxess Deutschland GmbH	2015	The inventive NBR examples are carried out in organic solvent solution polymerization, while water appears in the comparative emulsion process as the polymerization medium and later in coagulation/washing/workup.	Not disclosed in extracted evidence	Example 1: NBR was prepared in dimethylacetamide at 100°C with DoPAT modifier and 50% conversion, demonstrating organic-
EP2147933B1	LANXESS Inc.	2012	No water-related process step disclosed in the extracted evidence.	91%, 99.4%, 61%, 99%, and 85% hydrogenation levels in Examples 1-5	Example 1: Continuous hydrogenation of 2.5 wt.-% NBR in MCB at 130°C and 350 psi gave 91% hydrogenation.

US8362154B2	Lanxess Deutschland GmbH	2013	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Example 1.1: Hoveyda catalyst at 23 ppm Ru reduced Mw from 240 to 13 kg/mol within 425 min and gave gel-free degraded ni
US9593168B2	Arlanxeo Deutschland GmbH	2017	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Example 1.1: Hoveyda catalyst at 161.4 ppm Ru reduced Mw to 13 kg/mol by 425 min and produced gel-free degraded nitrile

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Hydrogenation-focused patents (EP3827029A1, US7666950B2, EP2473281B1, US9079979B2, EP2147933B1) disclose organic-solvent solution processing, most often in monochlorobenzene/chlorobenzene, whereas the precursor-manufacturing patent US9023914B2 uses organic-solvent polymerization for inventive NBR and water only in comparative emulsion examples.

- Direct HNBR hydrogenation patents report explicit hydrogenation metrics: US2147933B1 gives 61-99.4% hydrogenation levels, US7666950B2 reports residual double bonds from 0.9% to 25.4%, and EP2473281B1/US9079979B2 report RDB values down to 0.4% and 0.3% respectively.
- Catalyst systems differ substantially: EP3827029A1 centers on Ru/Pd/Rh hydrogenation catalysts; US7666950B2 and US9079979B2 use Grubbs/Hoveyda/Wilkinson-type catalysts in spinning-disk batch hydrogenation; EP2147933B1 uses osmium catalyst precursor in continuous flow; EP2473281B1 uses Ru/Os catalysts for simultaneous hydrogenation-metathesis.
- Process intensification is a major trend: US7666950B2 uses a spinning disk reactor, EP2147933B1 uses a multistage agitated continuous reactor, while EP2473281B1 and US9079979B2 combine hydrogenation with metathesis to reduce molecular weight in a single step.
- The degradation patents US8362154B2 and US9593168B2 do not directly report HNBR hydrogenation degree, but they show that catalyst choice and 1-hexene addition strongly affect molecular-weight reduction of NBR precursors, which is relevant to downstream HNBR manufacturing.

4. CONCLUSION

Across the selected primary patents, HNBR is predominantly produced by solution-phase hydrogenation of NBR using noble-metal catalysts, with the degree of hydrogenation controlled by catalyst identity, catalyst loading, pressure, temperature, and reactor design. The most directly informative patents report either residual double-bond content or hydrogenation percentage, showing that near-complete hydrogenation is achievable under optimized conditions, while lower catalyst loading or less favorable operating conditions leave substantially higher residual unsaturation. A second major theme is precursor tailoring by metathesis or controlled degradation to lower molecular weight before or during hydrogenation, enabling lower-viscosity HNBR grades. Water is generally absent from the HNBR hydrogenation medium and appears mainly in precursor emulsion polymerization or downstream workup in non-primary contexts.

5. REFERENCES

- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | ARLANXEO Deutschland GmbH | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>
- US7666950B2 | Process for preparing hydrogenated nitrile rubbers | Lanxess Deutschland GmbH | US | 2010 | <https://patents.google.com/patent/US7666950B2/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | LANXESS Deutschland GmbH | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>

- US9079979B2 | Process for the preparation of hydrogenated nitrile rubber | Lanxess Deutschland GmbH | US | 2015 | <https://patents.google.com/patent/US9079979B2/en>
- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Zannan Scitech Co., Ltd. | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>
- US9023914B2 | Nitrile rubbers and production thereof in organic solvents | Lanxess Deutschland GmbH | US | 2015 | <https://patents.google.com/patent/US9023914B2/en>
- EP2147933B1 | Process for the continuous hydrogenation of carbon-carbon double bonds in an ... | LANXESS Inc. | EP | 2012 | <https://patents.google.com/patent/EP2147933B1/en>
- US8362154B2 | Metathesis of nitrile rubbers in the presence of transition metal complex ... | Lanxess Deutschland GmbH | US | 2013 | <https://patents.google.com/patent/US8362154B2/en>
- US9593168B2 | Process for degrading nitrile rubbers in the presence of catalysts having an ... | Arlanxeo Deutschland GmbH | US | 2017 | <https://patents.google.com/patent/US9593168B2/en>